

Predictive Maintenance for Metro Climate and Air Systems: Checkpoint Report on Data Understanding, Pre-processing and Preliminary Modelling

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ABSTRACT

This checkpoint reports the completed data understanding and pre-processing stages, and preliminary clustering and anomaly-detection results, for the MetroPT-3 Air Production Unit dataset of 1,516,948 sensor readings. Three findings have changed the methodology set out in the proposal. Each is given here with the reasoning that produced the change, since in two of the three cases the reasoning is the more useful half.

First, the compression cycle failed as a unit of analysis. Segmentation recovered 10,974 load cycles, but only three of them begin inside a reported failure window, because during a failure the compressor stops cycling and runs continuously. The modelling unit is therefore now a fixed 15-minute window with cycle statistics carried in as features, which converts the disappearance of cycling from a hole in the data into a measurable signal. Second, unsupervised clustering on 41 window features selects $k = 2$ at silhouette 0.676 and recovers the fault regime without supervision: 98.5% of failure windows fall into a cluster holding 5.7% of the data. Third, and least expected, an Isolation Forest over all features scored worse than a single-feature duty threshold (F_1 0.312 against 0.577). Diagnosis attributes this to seasonal drift, with mean duty rising 50% from the training period to the test period, and to feature dilution; scoping the model to 14 physically motivated features recovered F_1 to 0.546.

The consequence is a deliberate narrowing of scope. Concurrent detection is effectively solved by physics and further model complexity buys almost nothing, so the remaining four weeks target the one open question: early warning. A leave-one-event-out experiment that removes all in-failure windows shows genuine but weak precursor signal, at 2.9x to 5.2x lift over base rate, carried principally by cycling frequency, which rises 20% in the 48 hours before onset while duty cycle stays flat.

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CCS Concepts

• **Computing methodologies** → Anomaly detection; Cluster analysis; • **Information systems** → Data stream mining; • **Applied computing** → Transportation.

Keywords

predictive maintenance, anomaly detection, sensor pre-processing, concept drift, class imbalance, railway systems

1 INTRODUCTION AND STATUS

The problem remains as stated in the proposal. A metro train's Air Production Unit compresses the air that drives climate control, doors, suspension levelling and braking assistance, so its degradation surfaces first as a comfort fault and finally as a train withdrawn mid-line. The MetroPT-3 record [12] provides 1,516,948 readings at 0.1 Hz over 213 days from an operational Porto Metro compressor, against four documented air-leak events.

Four of the eight scheduled weeks have elapsed; work began on 21 June 2027 and this checkpoint closes week 4 on 18 July. Data understanding, pre-processing, the feature store and clustering are complete; anomaly detection is in progress; early warning is open. This report covers what was built, what the data forced us to change, and how the remaining scope has been narrowed so that the project reaches a defensible finish line.

Three claims from the proposal have survived contact with the data, and one has not. Surviving: the duty cycle carries the fault signature; the label set is materially incomplete; per-sample accuracy is the wrong metric. Not surviving: the compression cycle as the unit of analysis. Section 5 explains why, because the reason is more useful than the correction.

2 RELATED WORK

The proposal grouped prior work into four lines; this section restates them briefly and marks where the checkpoint results have engaged with each, since several methods used below are drawn directly from them.

Physics-based diagnostics. Residual checks against an analytical model of the healthy system, with the ASHRAE RP-1043 chiller programme as the canonical reference [5], give interpretable alerts and need no labelled failures, but calibrate per unit and per duty profile. Our Section 6 result, in which a one-feature physical threshold outperformed a 41-feature learned model, vindicates the underlying instinct here. We had expected the opposite.

Deep anomaly detection on this APU. Davari et al. [6]



Figure 1: Data reduction through the pipeline. The row count rises at the resampling step because a uniform 10-second clock creates empty slots across the 909 hours of gaps in the record; those slots are flagged, never interpolated across.

applied sparse and LSTM autoencoders to this compressor, scoring reconstruction error over sliding windows. That work establishes the signal is learnable and sets a reference point. Our checkpoint finding that added capacity did not beat a physical threshold is the reason the LSTM comparison has been dropped from the remaining scope.

Streaming and drift-aware maintenance. Barros et al. [2] treat this unit in an operational streaming context, and Gama et al. [7] survey concept drift adaptation, with [8] covering the maintenance setting. This line proved the most directly applicable: the 50% rise in mean duty from our training window to the test period is a textbook instance of the drift these authors describe, and it is the diagnosed cause of our first model’s failure.

Evaluation under imbalance. Saito and Rehmsmeier [11] motivate precision-recall over ROC under heavy skew, which at 1.98% prevalence governs every metric we report; Bergmeir and Benítez [3] motivate blocked and chronological validation on temporally dependent data, which is why no random split appears anywhere in this work. Chandola et al. [4] supply the anomaly-detection taxonomy under which our task is a contextual, collective anomaly problem. Point-anomaly methods are the wrong family, and that classification is itself an argument for analysing windows.

The gap remains as stated. None of the work cited above treats the cyclic structure of the compressor as the carrier of the early signal, and we did not find the result of Section 7, that cycling frequency moves before duty cycle does, stated in that literature.

3 DATA UNDERSTANDING

3.1 Conditional distributions

The first substantive finding of the distributional analysis is that no analogue channel should be summarised unconditionally. TP2 has an overall mean of 1.37 bar, but that figure is a mixture over operating states and describes no state in particular: the channel sits near zero when the compressor is unloaded and rises sharply under load. Every summary in this report is therefore computed conditional on the digital state, and the modelling features record per-window minimum, maximum, mean and standard deviation, never a single central value.

The same logic exposed a redundancy. TP3 and Reservoirs correlate at $r = 1.00$ and PCA confirms an exactly zero-variance seventh component. We dropped one of the pair,

because collinear inputs split tree-based feature importance arbitrarily between two names for the same measurement, which would make the eventual attribution analysis uninterpretable.

3.2 Veracity: what the record does not contain

Profiling the timestamps established that 909.5 hours of the 213-day span are missing across 331 gaps, the longest 48 hours. That is 17.8% of calendar hours, a materially larger hole than the dataset documentation suggests, and it governs the pre-processing design in Section 4.

A second veracity question concerned frozen sensors. An initial count found 3.35% of raw rows bit-identical to their predecessor across all analogue channels. Treating that as a fault rate would have been an error: at 10-second resolution a quiet, idling machine legitimately repeats a reading. Requiring a sustained hold of at least 60 seconds on genuine acquisitions reduces the figure to 0.11%. The distinction matters because the first number would have justified discarding 3% of the record for no reason.

4 PRE-PROCESSING

Three decisions in this stage were non-obvious, and each was made against a specific property of this stream. None follows from a default.

4.1 Gap handling

Readings were snapped to a uniform 10-second grid. Holes shorter than 60 seconds are forward-filled, since the machine state cannot meaningfully change within six samples. Longer gaps are left null and marked with an explicit `is_gap` flag.

The reasoning is that interpolation across a 48-hour gap would manufacture smooth, well-behaved synthetic data precisely over the interval in which the unit was, in all likelihood, out of service. A model trained on that would learn the inverted lesson that the smoothest data is the most normal. Flagging costs 17.65% of grid rows, which is a different quantity from the 17.8% of calendar hours in Section 3.2, because the uniform 10-second grid creates empty slots across the gaps. Flagged rows are excluded from training but retained for audit.

4.2 Correct offsets only where zero is the expected rest value

All three of TP2, H1 and DV_pressure have minima below zero, which invites a blanket idle-referenced offset correction. Applying one is wrong. TP2 and DV_pressure rest at -0.012 and -0.020 bar, consistent with transducer zero-point bias. H1 rests at 8.644 bar, because at idle the cyclonic separator is genuinely holding pressure; H1 approaches zero only under load. An unguarded correction subtracts 8.644 bar from H1 and destroys the channel.

The implemented rule corrects a channel only when its idle median already lies within 0.5 bar of zero. H1 is rejected by that guard and left intact. This is recorded here because it is a general lesson about physically informed cleaning: the

statistic that identifies a bias in one channel identifies a signal in another, and only domain reasoning separates them.

4.3 Median filtering and debouncing

A five-sample rolling median suppresses transient spikes while preserving the sharp edges of state transitions. A mean filter was rejected because it smears exactly those edges, and the edges define the compression cycle boundaries that the feature extraction depends on. Digital channels are debounced with a 30-second minimum dwell, removing between 7 and 24,808 single-sample flickers per channel; without this, one flicker in COMP would split a single load cycle into three and corrupt every cycle-derived feature.

The complete pre-processing chain runs in 15.61 s over the full record, a throughput of approximately 97,000 readings per second.

5 THE CYCLE FAILED AS A UNIT OF ANALYSIS

The proposal committed to the compression cycle as the modelling grain, on the argument that a leak shows up in refill duration and inter-cycle interval. No instantaneous reading carries it. Segmentation on COMP transitions recovered 10,974 load cycles across the cleaned record, close to the expected order of magnitude.

Only three of those cycles begin inside a reported failure window.

The reason is physical and, in hindsight, obvious: during an air leak the compressor cannot restore reservoir pressure, so it stops cycling and runs continuously. Median cycle duration in normal operation is 140 s; the longest single continuous run inside a failure exceeds 91,000 s. The cycle is therefore not merely a poor unit inside the events, it effectively ceases to exist there, which means a cycle-indexed model would have almost no positive examples to learn from.

The correction was to demote the cycle from unit of analysis to feature source. The modelling unit is now a fixed 15-minute window, and cycle statistics (count, mean duration, maximum duration) enter as window features. This is strictly better than the original design for a reason worth stating: the window never disappears when the machine changes behaviour, so the collapse of cycling turns into a measurable feature instead of a hole. As Section 7 shows, that feature turns out to carry the earliest warning signal in the dataset.

Feature extraction produces 16,773 windows of 41 modelling features, plus an `n_readings` coverage count used to enforce the $\geq 50\%$ completeness filter and excluded from every model. Of the 16,773 windows, 332 (1.98%) fall inside a reported failure. The store is 5.99 MB and builds in 0.52 s.

6 PRELIMINARY MODELLING RESULTS

6.1 Clustering recovers the fault regime unsupervised

PCA retains 95% of variance in 15 components from 41 features. k -means was evaluated for $k \in [2, 8]$; the silhouette criterion [10] selects $k = 2$ decisively at 0.676, with Davies-Bouldin agreeing at 0.713, and both degrade sharply at $k \geq 3$.

Table 1 shows the result, and 98.5% of all failure windows fall in cluster 0. That an unsupervised method isolates the

Table 1: Cluster profile at $k = 2$. No label was used in fitting. Cluster 0 is the continuous-run regime.

	Cluster 0	Cluster 1
Share of windows	5.7%	94.3%
Mean duty fraction	0.997	0.109
Mean cycles per hour	0.004	0.686
Mean oil temperature	72.1 °C	62.1 °C
Mean motor current	5.53 A	1.84 A
Reported failure rate	34.0%	0.03%

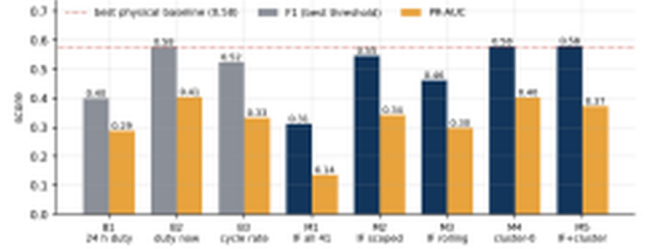


Figure 2: Test-period comparison (April to September). Grey bars are single-feature physical baselines; navy bars are learned models. The best learned result exceeds the best baseline by 0.003 F_1 .

fault regime without being told it exists is the strongest evidence available that the feature set encodes the underlying physics.

The caveat is equally important and quantifies a risk the proposal could only estimate: only 34.0% of cluster 0 is a reported failure. Continuous running is necessary but not sufficient for a reported fault. Either the operator’s reports are incomplete, or the metro genuinely demands sustained compressor output on some days. The data cannot adjudicate, and any precision figure in this report is capped by that ambiguity, whatever the model does.

6.2 The sophisticated model lost to the simple one

An Isolation Forest [9] was trained on February and March only, an interval with no reported failure, and used to score April through September. Evaluated at the window level on the test period, it achieved $F_1 = 0.312$ and PR-AUC 0.136. We report PR-AUC and not ROC-AUC throughout, because at 1.98% prevalence ROC curves are dominated by the true-negative mass [11].

A single-feature baseline, the instantaneous duty fraction, achieved $F_1 = 0.577$ and PR-AUC 0.405 on the same split. We could have retuned until the numbers improved and reported only that. Diagnosing the failure instead produced the most useful result of the stage. Two causes were identified. The first is concept drift [7]: mean duty fraction rises 50% from the February to March training window to the test period, so the detector rates ordinary later operation as anomalous relative to its reference and spends its alarm budget on seasonal change instead of on faults. Mean anomaly score on test normals is 0.447 against a training mean of 0.423, while test failures score 0.570; the failure signal is real but sits on an inflated floor. The second is feature dilution: Iso-

Table 2: Early-warning results, leave-one-event-out. Lift is PR-AUC divided by prevalence; a lift near 1.0 would mean nothing was learned.

H (h)	Prevalence	PR-AUC	Lift	Events flagged
6	0.58%	0.022	3.8x	4 / 4
12	1.17%	0.045	3.8x	4 / 4
24	1.98%	0.058	2.9x	4 / 4
48	4.03%	0.208	5.2x	4 / 4

lation Forest measures global rarity across all 41 dimensions, so rare-but-healthy states score as highly as faults.

Both diagnoses were tested. Scoping the model to 14 physically motivated features recovered F_1 to 0.546 and PR-AUC to 0.342, confirming dilution as the dominant cause. A rolling 30-day reference refitted weekly reached $F_1 = 0.463$, which is a smaller gain than expected and suggests the drift is partly within-reference and not purely between-period. Gating the scoped model by cluster membership reaches $F_1 = 0.580$, and that is the configuration we intend to carry forward. Figure 2 does not make it a clear winner. Cluster membership used on its own (M4) matches it at 0.58 and beats it on PR-AUC, 0.40 against 0.37, and the instantaneous-duty baseline (B2) also sits at 0.58, so three methods are tied on F_1 at the top of the figure. We prefer the gated model because cluster assignment is binary and therefore offers one operating point, whereas a continuous anomaly score can be traded against an alert budget. That is an argument about deployability and not about accuracy, and it is tested in week 7 rather than asserted here. If it does not hold, cluster-0 membership alone is the simpler system and should ship instead.

Figure 2 makes the summary uncomfortable: the best learned model beats the best one-line physical threshold by 0.003 F_1 . Concurrent detection on this unit is effectively solved by physics. Every method tested detects all four events; they differ only in false-alarm rate, between 69 and 235 alarm windows per 30 days.

7 THE OPEN QUESTION: EARLY WARNING

If detecting the fault while it is happening is trivial, the value is in predicting it beforehand. This was tested directly, and the experimental design matters more than the numbers.

All windows inside a failure were removed from both training and test, so the model cannot succeed by detecting the event it is meant to predict. Positive labels are the windows falling within H hours before onset. Validation is leave-one-event-out, holding out a 17-day span around each event in turn, because four events cannot support a random or even a single chronological split [3]. Because prevalence ranges from 0.58% at $H = 6$ to 4.03% at $H = 48$, raw PR-AUC is uninterpretable in isolation and is reported as lift over base rate.

Table 2 shows genuine but weak signal. The model beats chance at every horizon and flags all four events, but absolute precision remains too low to dispatch a technician on. The 48-hour horizon performs best, consistent with the precursor analysis in the proposal, which found lead times of 36 to 62



Figure 3: Three-stage signature. Cycling frequency rises before onset and collapses at onset; duty cycle does not move until onset.

Table 3: Measured performance over the full record, single machine. Rows are additive; the six pre-processing stages sum to the 15.61 s quoted in Section 4.3.

Stage	Time
CSV load (1,516,948 rows)	5.19 s
Resample, gap flagging	7.18 s
Offset correction	0.12 s
Median filtering, 7 channels	2.09 s
Digital debouncing, 8 channels	0.60 s
Frozen-channel detection	0.43 s
Window feature extraction	0.52 s
k -means, $k = 2$	0.07 s
DBSCAN	5.63 s
Isolation Forest training	0.26 s
Isolation Forest scoring, all windows	0.13 s
Pre-processing throughput	97,190 rows/s
Per-window scoring latency	0.008 ms
Rolling refit, 18 weekly refits	2.2 s

hours for two of the four events.

7.1 A three-stage degradation signature

The feature separation at $H = 48$ explains where the early signal lives, and it was not where the proposal expected. Mean cycling rate is 2.60 per hour in healthy operation, 3.11 per hour in the 48 hours before a failure, a 20% rise, and 0.13 per hour inside a failure, a 95% collapse. Duty fraction, by contrast, is 0.143 healthy and 0.145 pre-failure, moving only at onset to 0.988.

This is physically coherent. A developing leak drains the reservoir faster, so the unit refills more often, producing more frequent short cycles. Once the leak outpaces compressor capacity, cycling ceases and the unit runs flat out. The operational implication is sharp: any duty-based alarm has essentially zero lead time by construction, because duty does not move until the fault is already present. The early-warning model must be built on cycle-frequency features, and the two strongest univariate separators at $H = 48$ are indeed trailing 1-hour and 6-hour cycle counts.

8 EFFICIENCY

Efficiency was measured because a monitoring system that cannot keep pace with its own data stream is not deployable, and because 1.5 M points is large enough for the question to be real.

Table 3 shows the deployment question is settled comfortably. Per-window scoring latency, the 0.13 s scoring row divided across the 16,773 windows, is 0.008 ms, roughly eight orders of magnitude below the 15-minute window period. Charged end to end, including pre-processing, the figure is 0.97 ms per window, about six orders of magnitude, so the online scorer is not remotely a bottleneck. Weekly drift-adaptive refitting costs 2.2 s in total across the whole test period, which removes any efficiency argument against re-training often. The binding constraint is resampling and gap flagging at 7.18 s, with CSV parsing close behind at 5.19 s; both are ingestion costs, and a columnar store removes most of them in production.

9 CHALLENGES AND BACKUP PLAN

9.1 Current roadblocks

Label incompleteness is now measured rather than estimated. Two thirds of the continuous-run cluster carries no maintenance report. Reported precision is therefore bounded by the label set, whatever the models do, and no amount of tuning will lift it. Mitigation is procedural: unlabelled high-score episodes are reported as a separate category with their sensor signatures characterised against confirmed events.

Four events is a hard statistical floor. Every supervised figure rests on four positive episodes. We use leave-one-event-out throughout and report spread alongside every mean.

Concept drift has already broken one model [7]. The 50% rise in mean duty from training to test period is the direct cause of the Isolation Forest’s failure. Rolling reference estimation is implemented and costs 2.2 s.

Early-warning precision may not reach a usable level. Lift is real at 2.9x to 5.2x but absolute precision is low at every horizon tested.

One result we cannot yet explain. The rolling 30-day reference was expected to be the strongest of the three drift mitigations and was the weakest, at $F_1 = 0.463$ against 0.546 for the same model on a fixed reference. If the problem were purely between-period drift, the rolling version should have won. We have offered within-reference drift as an explanation but have not tested it, and the refit interval and window length are both unvalidated choices. This is listed here as an open question, not as a finding.

9.2 Backup plan and switch rule

If the early-warning model does not reach usable precision, the fallback is online change-point detection [1] on the univariate cycle-rate series. Note that the fallback has changed since the proposal, which specified the duty series; Section 7 shows duty carries no pre-onset signal, so a change-point detector on duty would inherit zero lead time. Cycle rate is the correct univariate carrier.

A stated rule governs the switch, so that the decision is not made by judgement under deadline pressure: if leave-one-event-out PR-AUC lift at the 24-hour horizon remains below 3x at the end of week 6 (1 August), tuning stops and

Table 4: Remaining schedule, 19 July to 15 August 2027.

Week	Dates	Activity
5	19–25 Jul	Early-warning features: cycle-frequency and pressure-recovery statistics targeted at the pre-failure stage
6	26 Jul–1 Aug	Gradient boosting at 24 h and 48 h horizons; apply the switch rule
7	2–8 Aug	Efficiency benchmarking, feature ablations, sensitivity to alert budget
8	9–15 Aug	Final report, reproducible repository, dashboard mock

the change-point fallback ships. A backup plan without a trigger condition is not a plan.

10 REVISED SCOPE AND TIMELINE

The remaining four weeks are deliberately narrowed to the open question.

One item has been dropped: the LSTM autoencoder. Isolation Forest already demonstrated that added model capacity does not beat a physical threshold on this signal, and the diagnosis attributed that to drift and dilution rather than to insufficient capacity. A deep model would consume a week to answer a question already answered. Dropping it is what makes the remaining schedule realistic.

11 CONCLUSION

Halfway through, the pipeline reduces 1,516,948 raw readings to 16,773 analysable windows in about 16 seconds, and unsupervised clustering recovers the fault regime with no labels. The three results that matter are that the compression cycle failed as a modelling unit for a physically interesting reason and was successfully repurposed as a feature source; that concurrent fault detection is solved by a single physical feature and learned models add essentially nothing to it; and that the genuine open problem is early warning, where signal exists at 2.9x to 5.2x lift and is carried by cycling frequency. Duty cycle contributes nothing before onset.

Keeping the big picture in view: none of these tasks is an end in itself. The outcome that matters is whether a maintenance planner receives enough warning to pull a train before it strands passengers mid-line. Concurrent detection does not deliver that, which is precisely why the remaining scope targets lead time.

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